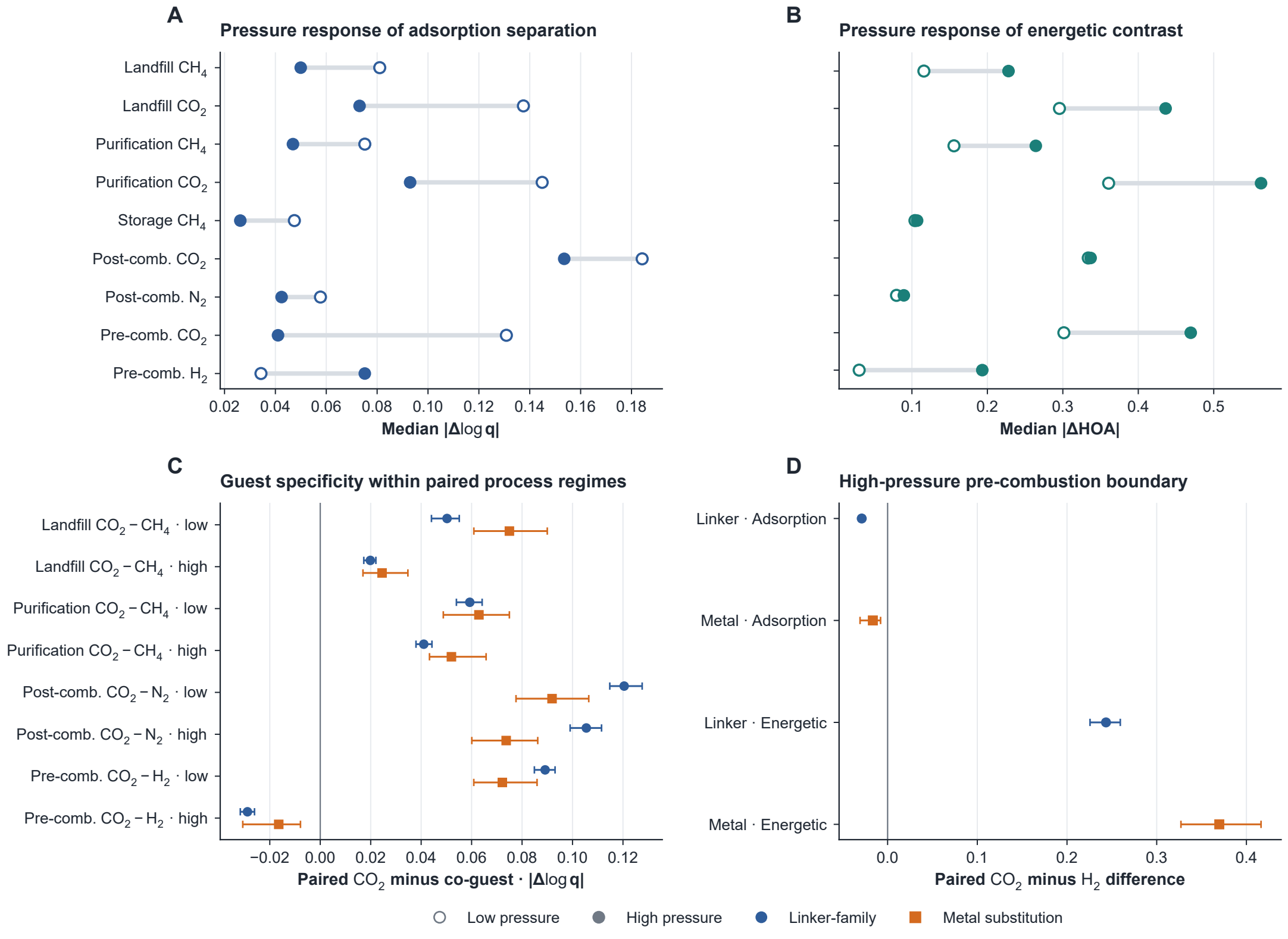

A
Pressure response of adsorption separation
Landfill CH4
Landfill CO2
Purification CH4
Purification CO2
Storage CH4
Post-comb. CO2
Post-comb. N2
Pre-comb. CO2
Pre-comb. H2
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
0.18
Median |Δlog q|
B
Pressure response of energetic contrast
0.1
0.2
0.3
0.4
0.5
Median |ΔHOA|
C
Guest specificity within paired process regimes
Landfill CO2 − CH4 · low
Landfill CO2 − CH4 · high
Purification CO2 − CH4 · low
Purification CO2 − CH4 · high
Post-comb. CO2 − N2 · low
Post-comb. CO2 − N2 · high
Pre-comb. CO2 − H2 · low
Pre-comb. CO2 − H2 · high
−0.02
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Paired CO2 minus co-guest · |Δlog q|
D
High-pressure pre-combustion boundary
Linker · Adsorption
Metal · Adsorption
Linker · Energetic
Metal · Energetic
0.0
0.1
0.2
0.3
0.4
Paired CO2 minus H2 difference
Low pressure
High pressure
Linker-family
Metal substitution